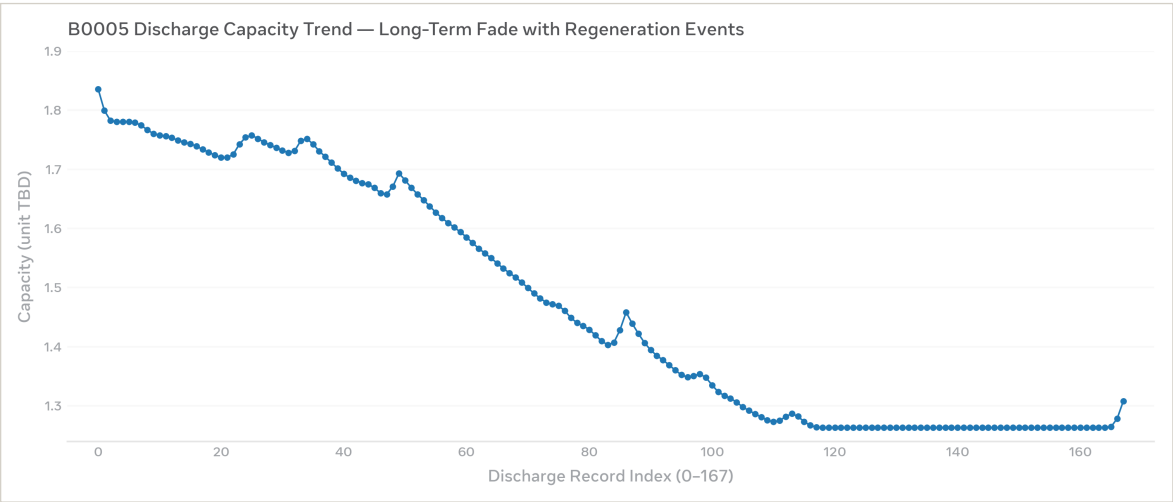


QUESTION Can one capacity number be treated as degradation?



Discharge capacity vs. index. Five upward jumps on the fade curve. NASA PCoE B0005.

KEY NUMBERS

168 discharges · 1.86 → 1.33 (–29%) · 5 jumps · largest +0.09 · 24 °C at all five
Impedance also rises overall and is not monotonic. Time gaps do not follow one pattern.

WHAT THE DATA SHOWS

Laboratory cell, 168 discharges. Long-term fade is real. Short-term capacity can jump up, then fall again. Ambient temperature does not explain the jumps.

JUDGMENT

A single capacity point is not the irreversible state at that moment.
Health = trend + measurement context. Not one absolute.

BESS RULE

Do not treat every SOH uptick as recovery or every drop as faster aging. Check last full cycle, rest, diagnostic history, temperature, operating condition, and whether the change persists.

WHAT THIS DOES NOT PROVE

Not a single mechanism. Not field magnitude. Not a plant diagnosis.

WHY THIS MATTERS FOR BESS

B0005 is not a plant. The transferable lesson is the failure mode:
a monitoring stack that treats every SOH point as truth will
raise false recovery and false aging alarms.

Use trend, context, and persistence before you change operations.

WHEN TO USE THIS REVIEW

When EMS still looks normal,
but SOH or capacity just jumped.

bessdiag.com
github.com/Anthoony2023/bess-diagnostics